

3-D Printing of Open Source Appropriate Technologies for Self-Directed Sustainable Development

Authors: Joshua M. Pearce, et al. (2010)

ABSTRACT

An extensive study validating 3D printing as a viable method to manufacture high-quality, open-source appropriate technologies at a fraction of standard commercial costs, encouraging localized development and customization.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides the socio-economic and technical justification for utilizing 3D-printed structural links in the EB15 prototype.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Joshua M. Pearce provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.